

Lab T4 - Personal Report

Pham Van Hoang Anh Tu - MSSV chua duoc cung cap

1. Problem

Camera-LiDAR timestamp offset creates position misalignment in ADAS fusion. LiDAR captures at t_{ref} minus offset while fusion uses the ego pose at t_{ref} . This is a synthetic 2-D alignment benchmark.

2. Method

Park et al. (RA-L 2020) estimates camera-LiDAR extrinsic and time lag. Huai et al. (2021) studies rolling-shutter camera-IMU calibration. The group uses a simpler 2-D proxy benchmark, not a reproduction of either paper.

3. Benchmark

The group ran 45 conditions and 4,500 samples. All 0 ms baselines are 0.023-0.027 m. A at 100 ms and d=20 m gives $E_{pre}=2.001$ m and linear $E_{post}=0.026$ m.

4. Failure case

C at d=40 m and offset=100 ms has linear $E_{post}=1.351$ m, above the group-defined 0.5 m threshold. CTRV gives 0.025 m on the same samples.

5. Engineering decision

Use CTRV with a reliable offset and yaw rate. If offset exceeds 30 ms and yaw rate exceeds 0.2 rad/s, avoid LiDAR-to-camera-bbox association, track independently, and log a synchronization warning.

Evidence: results/results.csv, results/samples.csv, results/run_log.txt, plots/error_vs_offset.png, plots/alignment_turn.png. Command: `python -m src.run_benchmark --config config_chot.yaml`.